

Definitions of functional groups and other named substructures in CHMTRN and suggestions for them in SLICE

version 1.2

Written by P N Judson with acknowledgement to O Grushin for significant input and advice

This document contains substructural diagrams of the named functional groups and substructures in CHMTRN/PATRAN, along with qualifying statements where necessary, and suggestions for their implementation in SLICE. Some are unchanged but additional constraints are suggested for others. The order in which they are listed may seem odd – it is the order used in the CHMTRN/PATRAN group table for technical reasons.

There are multiple patterns in the functional group definitions because slightly different ones are sometimes needed in CHMTRN mechanism “ATTACH” and “INTRODUCE” statements from the ones used in qualifying statements. For example, in a qualifying statement, “ETHER” means anything of the form R-O-R whereas in an ATTACH statement, it means “attach OMe”. There are no similar mechanism statements in SLICE because transformations are entered graphically. So the PATRAN definitions have been interpreted using only the patterns intended for use in qualifying statements.

Where H is used in this document it means it must be present on the atom it is attached to in a target structure, whether explicitly or implicitly.

I have assumed that atom connectivity in SLICE means the number of attached atoms including hydrogen. Please correct me if I am wrong.

I have also assumed that specifying, e.g., connectivity = 2 for the sulfur in R-S-R has the same effect as specifying there must be two electron pairs on the sulfur atom (you cannot specify connectivity as an atom property in PATRAN). Is my assumption correct? As long as it is, wherever atom pair count is mentioned in the SLICE column in the tables it can be dropped and connectivity can be used instead. However, if connectivity refers only to non-hydrogen atoms (or those plus explicit hydrogens), electron pair count is necessary. In the example here, R-S-R the electron pair count is used in PATRAN to restrict matching to thioethers, excluding things like sulfones and sulfoxides.

The functional group origin, which is always a carbon atom, is marked with an asterisk (*).

Functional groups

ALLENE

CHMTRN/PATRAN	SLICE	Comments
$\text{*C}=\text{C}=\text{C}$ not aryl C1, C3	$\text{*C}=\text{C}=\text{C}$ not aryl C1, C3	Aryl ought to be chemically impossible. My guess is that the constraint was included in case aromatic perception uses a simple version of the Hückel rule leading to an equivalent of the well-known error for, e.g., hexamethylene cyclohexane. So it has been kept in the SLICE definition.

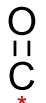
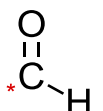
OLEFIN

CHMTRN/PATRAN	SLICE	Comments
$\text{*C}=\text{C}$ not allene, not aryl	$\text{*C}=\text{C}$ not allene, not aryl, no hetero atoms alpha to either carbon	The PATRAN definition is too vague. An enol, for example, is so different that it clearly should be excluded.

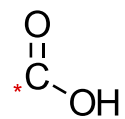
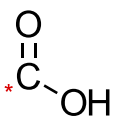
KETONE

CHMTRN/PATRAN	SLICE	Comments
$\begin{array}{c} \text{O} \\ \\ \text{C} \\ \text{*} \end{array}$ O not in a ring and no alpha hetero; no H, no additional alpha hetero, and no additional double bond, on C	$\begin{array}{c} \text{O} \\ \\ \text{C} \\ / \quad \backslash \\ \text{C} \quad \text{C} \end{array}$ O connectivity = 1	I.e. no change, assuming the CHMTRN restriction on the O was intended to exclude compounds with an additional carbon alpha to the oxygen such as pyrylium (PATRAN did not have connectivity as an attribute that you could specify)

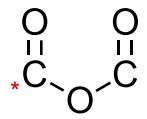
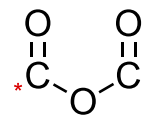
ALDEHYDE

CHMTRN/PATRN	SLICE	Comments
 O not in a ring and no alpha hetero; at least one H, no additional alpha hetero, and no additional double bond, on C	 Exactly one hetero atom alpha to C*; O connectivity = 1	The exclusion of the additional double bond in the CHMTRN is redundant, since the presence of at least one H is required.

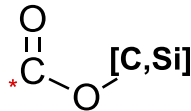
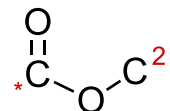
ACID (Carboxylic acid in SLICE)

CHMTRN/PATRN	SLICE	Comments
 Carbonyl O not in a ring and no hetero; C* not in a ring	 Exactly two hetero atoms alpha to C*; carbonyl O connectivity = 1	The CHMTRN definition allows an extra hetero atom on the carbon. So the pattern matches for, e.g., carbamic acids, $\text{NR}_2\text{-C(=O)-OH}$. The difference in their reactivity is too great to include them,

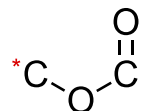
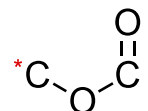
ANHYDRIDE

CHMTRN/PATRN	SLICE	Comments
	 Exactly 2 hetero atoms alpha to each of the carbon atoms	More restrictive in SLICE Allowing additional heteratoms alpha to the terminal carbons would cause matches to weird things such as $\text{Me-C(=O)-O-C(=O)-NR}_2$

ESTER

CHMTRN/PATRAN	SLICE	Comments
 Not anhydride on C*	 Exactly 2 hetero atoms alpha to C* If C2 is not aromatic then exactly one hetero atom alpha to it	More restrictive in SLICE If we do want to cover silyl ester, I think it should be a group in its own right

ESTERX

CHMTRN/PATRAN	SLICE	Comments
 C* not ester, acid, or anhydride; carbonyl carbon must be ester	 If C* is not aromatic then exactly one hetero atom alpha to it; exactly 2 hetero atoms alpha to the carbonyl carbon	More restrictive in SLICE

ACID*HALIDE

CHMTRN/PATRAN	SLICE	Comments
<chem>*C(=O)[Cl,Br,I]</chem>	<chem>1C-C(=O)*[Cl,Br,I]</chem> or <chem>2C-O-C(=O)*[Cl,Br,I]</chem> No heteroatoms on C2 if it is not aromatic	I have assumed that it is all right to specify more than one pattern Alkyl haloformates have similar reactivity to acid halides and so they have been included but dialkylchloroformamides, which have different reactivity, have not.

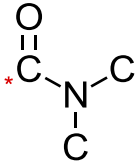
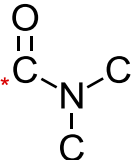
AMIDE*1

CHMTRN/PATRAN	SLICE	Comments
<chem>*C(=O)NH2</chem> 2 or 3 hetero atoms alpha to the carbon	<chem>*C(=O)NH2</chem> Exactly 2 hetero atoms alpha to the carbon	More restrictive in SLICE

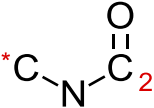
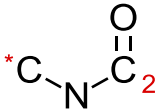
AMIDE*2

CHMTRN/PATRAN	SLICE	Comments
<chem>*C(=O)NC</chem> 2 or 3 hetero atoms alpha to C*	<chem>*C(=O)NC2</chem> Exactly 2 hetero atoms alpha to C*; If C2 is not aromatic then exactly 1 hetero atom alpha to it	More restrictive in SLICE

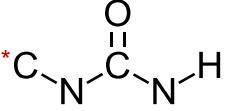
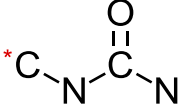
AMIDE*3

CHMTRN/PATRAN	SLICE	Comments
 2 or 3 hetero atoms alpha to the carbon	 Exactly 2 hetero atoms alpha to C*; For each of the other two carbon atoms: if it is no aromatic then exactly 1 hetero atom alpha to it	More restrictive in SLICE

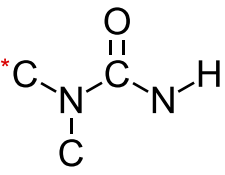
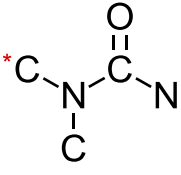
AMIDZ

CHMTRN/PATRAN	SLICE	Comments
 must be amide on carbon 2 no hetero atom alpha to the nitrogen	 If C* is not aromatic then exactly 1 hetero atom alpha to it exactly 2 hetero atoms alpha to carbon 2; no hetero atom alpha to the nitrogen	Slightly more restrictive in SLICE

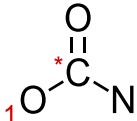
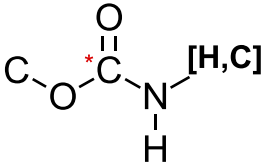
N*UREA*H

CHMTRN/PATRAN	SLICE	Comments
 C* not amide*2, not amide*3; no hetero atom alpha to either of the nitrogen atoms	 Exactly one hetero atom alpha to C*; no hetero atom alpha to either of the nitrogen atoms	Slightly more restrictive in SLICE

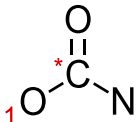
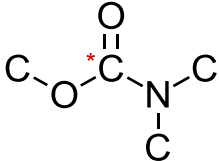
N*UREA*C

CHMTRN/PATRAN	SLICE	Comments
 C* not amide*3; no hetero atom alpha to either of the nitrogen atoms	 No hetero atom alpha to either of the nitrogen atoms; For each carbon atom except the carbonyl one: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

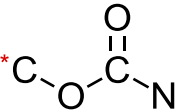
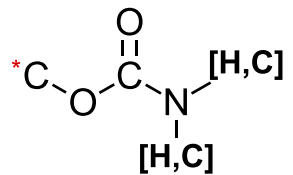
CARBAMATE*H

CHMTRN/PATRAN	SLICE	Comments
 No H and no hetero atom alpha to O1; 1 or 2 H alpha to the nitrogen; no hetero atom alpha to the nitrogen	 For each carbon atom except the carbonyl one: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

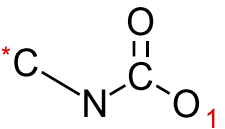
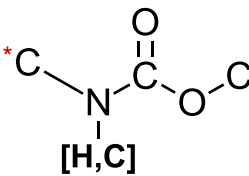
CARBAMATE*C

CHMTRN/PATRAN	SLICE	Comments
 No H and no hetero atom alpha to O1; no H and no hetero atom alpha to the nitrogen	 For each carbon atom except the carbonyl one: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

O*CARBAMATE

CHMTRN/PATRAN	SLICE	Comments
 No heteroatoms alpha to the nitrogen	 For each carbon atom except the carbonyl one: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

N*CARBAMATE

CHMTRN/PATRAN	SLICE	Comments
 No H and no hetero atom alpha to O1; no hetero atom alpha to the nitrogen	 For each carbon atom except the carbonyl one: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

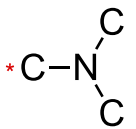
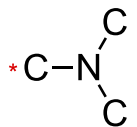
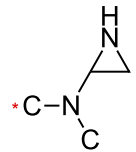
O*CARBONATE

CHMTRN/PATRAN	SLICE	Comments
 No H and no hetero atom alpha to O1; no ester, acid, or anhydride on C*	 For C* and C2: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

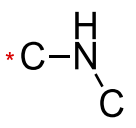
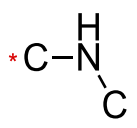
AZIRIDINE

CHMTRN/PATRAN	SLICE	Comments
 ?	 Exactly one hetero atom alpha to each of the carbon atoms	There appears to be an error in the PATRAN definition of aziridine. From comparison with, e.g., epoxide and from usage of "AZIRIDINE" in some transforms, the definition for SLJICE appears to be what was intended

AMINE*3

CHMTRN/PATRAN	SLICE	Comments
 No amide and no aziridine on any of the carbon atoms	 For each carbon atom: not in a common ring with the nitrogen of size < 4; if the carbon atom is not aromatic then not multiply bonded and exactly one hetero atom alpha to it	Slightly more restrictive in SLICE The exclusion of aziridine in the PATRAN definition is probably to exclude the cases where two of the carbon atoms and the nitrogen form an aziridine. It is not clear whether exclusion of something like  was intended or was an unintended side effect.

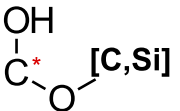
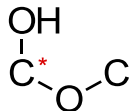
AMINE*2

CHMTRN/PATRAN	SLICE	Comments
 No amide and no aziridine on either of the carbon atoms. No hetero atom alpha to the nitrogen atom.	 No hetero atom alpha to the nitrogen atom For each carbon atom: if the carbon atom is not aromatic then not multiply bonded and exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

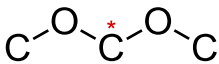
AMINE*1

CHMTRN/PATRAN	SLICE	Comments
*C-NH_2 No amide and no aziridine on the carbon atom; no hetero atom alpha to the nitrogen atom	*C-NH_2 if the carbon atom is not aromatic then not multiply bonded and exactly one hetero atom alpha to it; no hetero atom alpha to the nitrogen atom	Slightly more restrictive in SLICE

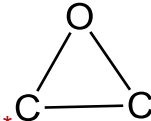
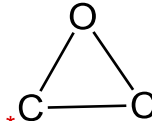
HEMIACETAL in CHMTRN – **rename to hemiacetal-ketal in SLICE**

CHMTRN/PATRAN	SLICE	Comments
 No ester, acid, or anhydride on C*; no charge on the ether oxygen	 Exactly 2 hetero atoms alpha to C*; If the other carbon atom is not aromatic then exactly one hetero atom alpha to it	I suggest that if we want to allow silicon it should be defined as a separate functional group. Agree.

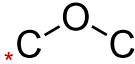
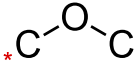
ACETAL – **rename to acetal-ketal in SLICE**

CHMTRN/PATRAN	SLICE	Comments
 No ester or anhydride on C* No ester and no anhydride on C*; no charge on and no hetero atom alpha to either oxygen; either oxygen, but not both, allowed to be in a ring size > 3	 Exactly 2 hetero atoms alpha to C*; For each of the other two carbon atoms: if the carbon atom is not aromatic then exactly 1 hetero atom alpha to it	I suggest that if we want to allow silicon we should define separate functional groups.

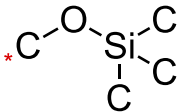
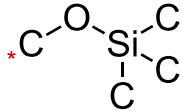
EPOXIDE

CHMTRN/PATRAN	SLICE	Comments
 No acetal, hemiacetal, ester, or anhydride on either carbon atom	 Exactly 1 hetero atom alpha to each carbon atom	Slightly more restrictive in SLICE

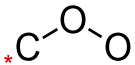
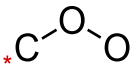
ETHER

CHMTRN/PATRAN	SLICE	Comments
 No acetal, hemiacetal, ester, or anhydride on either of the carbon atoms; oxygen allowed to be in a ring but only if it is size > 3	 If C* is not aromatic then not multiply bonded and exactly 1 hetero atom alpha to it; if the other C is not aromatic then exactly 1 hetero atom alpha to it	Slightly more restrictive in SLICE

TRIALKYLSELOXY

CHMTRN/PATRAN	SLICE	Comments
 Exactly 1 hetero atom alpha to the silicon	 If C* is not aromatic then exactly 1 hetero atom alpha to it; exactly 1 hetero atom alpha to each of the other carbon atoms	Slightly less restrictive in SLICE

PEROXIDE

CHMTRN/PATRAN	SLICE	Comments
 Exactly 1 hetero atom alpha to each oxygen	 Exactly 1 hetero atom alpha to the terminal oxygen	I.e. no change (There would be no harm in excluding additional hetero atoms alpha to the middle oxygen, as in PATRAN, but it is not needed – there will not be cases in practice)

ALCOHOL

CHMTRN/PATRAN	SLICE	Comments
C-OH No acetal, hemiacetal, or acid on C	*C-OH Exactly 1 hetero atom alpha to C*; If C* is not aromatic then C* not multiply bonded	More restrictive in SLICE we would not want, e.g., C(=S)-OH to be perceived as an alcohol

NITROSO

CHMTRN/PATRAN	SLICE	Comments
*C-N=O Exactly one hetero atom alpha to the nitrogen and one hetero atom alpha to the oxygen; no H alpha to the nitrogen or oxygen; neither the nitrogen nor the oxygen allowed to be in a ring	*C-N=O Exactly one hetero atom alpha to the nitrogen and one hetero atom alpha to the oxygen; no H on the nitrogen or oxygen; neither the nitrogen nor the oxygen allowed to be in a ring	I.e. no change

DIAZO

CHMTRN/PATRAN	SLICE	Comments
*C=N=N- Exactly 1 hetero atom alpha to each nitrogen atom and no H alpha to either of them; neither nitrogen in a ring	*C=N=N- (Exactly 1 hetero atom alpha to each nitrogen atom and no H alpha to either of them; neither nitrogen in a ring?)	I.e. no change In practice, the nitrogens cannot have other alpha atoms or be in rings.

HYDRAZONE

CHMTRN/PATRAN	SLICE	Comments
*C=N-N Exactly 1 hetero atom alpha to each nitrogen; no H on the middle nitrogen and not in a ring	*C=N-N Exactly 1 hetero atom alpha to each nitrogen; no H on the middle nitrogen and not in a ring	I.e. no change

OXIME

CHMTRN/PATRAN	SLICE	Comments
*C=N-OH Exactly 1 hetero atom alpha to the nitrogen and no hydrogen	*C=N-OH Exactly 1 hetero atom alpha to the nitrogen and no hydrogen	I.e. no change

IMINE

CHMTRN/PATRAN	SLICE	Comments
*C=N	*C=N No hetero atoms alpha to C*; no hetero atoms alpha to the nitrogen	More restrictive in SLICE (In PATRAN, oxime etc would be excluded anyway because they are defined as functional groups in their own right)

THIOCYANATE

CHMTRN/PATRAN	SLICE	Comments
2 electron pairs and no H on the sulfur	If C* is not aromatic then exactly 1 hetero atom alpha to it; 2 electron pairs on the sulfur (or sulfur connectivity = 2)	Slightly more restrictive in SLICE

ISOCYANIDE

CHMTRN/PATRAN	SLICE	Comments
	If C* is not aromatic then exactly 1 hetero atom alpha to it	Slightly more restrictive in SLICE

NITRILE

CHMTRN/PATRAN	SLICE	Comments
Exactly 1 hetero atom alpha to C*	Exactly 1 hetero atom alpha to C*	I.e. no change

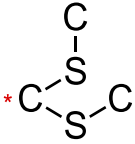
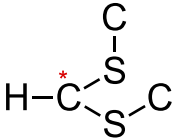
ISOCYANATE

CHMTRN/PATRAN	SLICE	Comments
$\text{*C}-\text{N}=\text{C}=\text{O}$ No H and no hetero atom alpha to the oxygen or nitrogen or the carbon between them and none of them in rings	$\text{*C}-\text{N}=\text{C}=\text{O}$ if C* is not aromatic then exactly 1 hetero atom alpha to it	Slightly more restrictive in SLICE (the constraint “No H and no hetero atom alpha to the oxygen or nitrogen or the carbon between them and none of them in rings” does not seem to me to be needed)

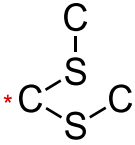
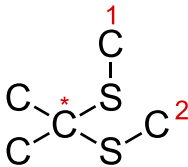
THIOESTER

CHMTRN/PATRAN	SLICE	Comments
$\begin{array}{c} \text{O} \\ \\ \text{*C}-\text{S}-\text{C} \end{array}$ No H and no hetero atom alpha to, and no ring on, the oxygen; 2 electron pairs on the sulfur; no H and no hetero atom alpha to the sulfur	$\begin{array}{c} \text{O} \\ \\ \text{*C}-\text{S}-\text{C} \end{array}$ Exactly 2 hetero atoms alpha to C*; 2 electron pairs on the sulfur (or sulfur connectivity = 2); oxygen connectivity = 1; if the other carbon is not aromatic then exactly 1 hetero alpha to it	Slightly more restrictive in SLICE

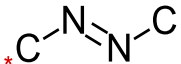
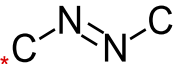
DITHIOACETAL

CHMTRN/PATRAN	SLICE	Comments
 At least 1 H alpha to C*; no H and no hetero atom alpha to either of the sulfur atoms; 2 electron pairs on each sulfur	 Exactly 2 hetero atoms alpha to C*; 2 electron pairs (or sulfur connectivity = 2) on each sulfur	Slightly more restrictive in SLICE Dithioacetal and dithioacetal are used only in early Harvard transforms, pre-pattern chemistry. So we could merge them without creating problems or drop them altogether.

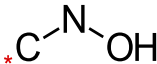
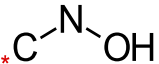
DITHIOKETAL

CHMTRN/PATRAN	SLICE	Comments
 No H alpha to C*; no H and no hetero atom alpha to either of the sulfur atoms; 2 electron pairs on each sulfur	 Exactly 1 hetero atom alpha to C1; exactly 1 hetero atom alpha to C2; 2 electron pairs (or sulfur connectivity = 2) for each sulfur	Slightly more restrictive in SLICE See comments on dithioacetal

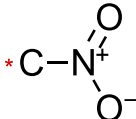
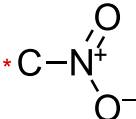
AZO

CHMTRN/PATRAN	SLICE	Comments
 No H on either nitrogen; exactly 2 hetero atom alpha to each nitrogen	 No H on either nitrogen; exactly 1 hetero atoms alpha to each nitrogen	I.e. no change

HYDROXYLAMINE

CHMTRN/PATRAN	SLICE	Comments
 Exactly 1 hetero atom alpha to the nitrogen and 1 hetero atom alpha to the oxygen	 Exactly 1 hetero atom alpha to the nitrogen and 1 hetero atom alpha to the oxygen	I.e. no change

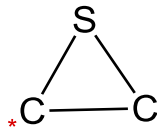
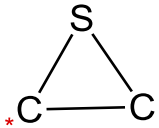
NITRO

CHMTRN/PATRAN	SLICE	Comments
		I.e. no change

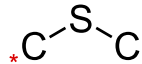
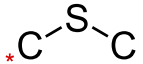
THIOL

CHMTRN/PATRAN	SLICE	Comments
*C-SH No hetero atom alpha to the sulfur atom; 2 electron pairs on the sulfur atom	*C-SH If C* is not aromatic then C* not multiply bonded; exactly 1 hetero atom alpha to C*; 2 electron pairs on the sulfur (or sulfur connectivity = 2)	Slightly more restrictive in SLICE

EPISULFIDE

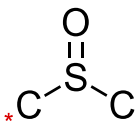
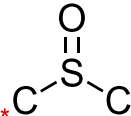
CHMTRN/PATRAN	SLICE	Comments
 No H and no hetero atom alpha to the sulfur; no thioester on either of the carbon atoms	 2 electron pairs on the sulfur (or sulfur connectivity = 2); exactly 1 hetero atom alpha to each of the carbon atoms	Slightly more restrictive in SLICE

SULFIDE

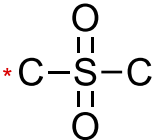
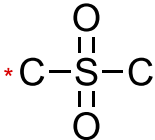
CHMTRN/PATRAN	SLICE	Comments
 No H and no hetero atom alpha to the sulfur; 2 electron pairs on the sulfur; no episulfide, thioester, or thiocyanate on either of the carbon atoms	 2 electron pairs on the sulfur (or sulfur connectivity = 2); For each carbon atom:	Slightly more restrictive in SLICE

	if the carbon atom is not aromatic then exactly one hetero atom alpha to it	
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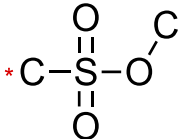
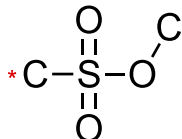
SULFOXIDE

CHMTRN/PATRAN	SLICE	Comments
 Exactly 1 hetero atom alpha to the sulfur atom; exactly 1 hetero atom alpha to the oxygen atom; no H on the sulfur atom or the oxygen atom; 1 electron pair on the sulfur atom	 1 electron pair on the sulfur (or sulfur connectivity = 3); For each carbon atom: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

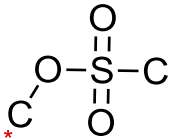
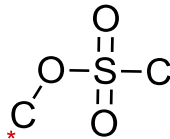
SULFONE

CHMTRN/PATRAN	SLICE	Comments
 No H on the sulfur atom; exactly 2 hetero atoms alpha to the sulfur atom; no H on the oxygen atoms and no rings; exactly 1 hetero atom alpha to each oxygen atom	 No H on the sulfur atom; exactly 2 hetero atoms alpha to the sulfur atom; no H on the oxygen atoms and no rings; exactly 1 hetero atom alpha to each oxygen atom; For each carbon atom: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

C*SULFONATE

CHMTRN/PATRAN	SLICE	Comments
 Exactly 3 hetero atoms alpha to the sulfur atom; no H on the double-bonded oxygen atoms and no rings; exactly 1 hetero atom alpha to each of the three oxygen atoms;	 Exactly 3 hetero atoms alpha to the sulfur atom; no H on the double-bonded oxygen atoms and no rings exactly 1 hetero atom alpha to each of the three oxygen atoms; For each carbon atom: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

O*SULFONATE

CHMTRN/PATRAN	SLICE	Comments
 Exactly 3 hetero atoms alpha to the sulfur atom; no H on the double-bonded oxygen atoms and no rings; Exactly 1 hetero atom alpha to each of the three oxygen atoms	 Exactly 3 hetero atoms alpha to the sulfur atom; no H on the double-bonded oxygen atoms and no rings; For each carbon atom: if the carbon atom is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE

ACETYLENE

CHMTRN/PATRAN	SLICE	Comments
$^* \text{C} \equiv \text{C}$	$^* \text{C} \equiv \text{C}$ No heteroatoms alpha to either carbon	I.e. no change

PHOSPHINE

CHMTRN/PATRAN	SLICE	Comments
$ \begin{array}{c} [\text{H}, \text{C}] \\ \\ \text{C} - \text{P} \\ \quad \\ ^* \quad [\text{H}, \text{C}] \end{array} $ No H and no hetero atom on the phosphorus atom; 1 electron pair on the phosphorus atom	$ \begin{array}{c} [\text{H}, \text{C}] \\ \\ \text{C} - \text{P} \\ \quad \\ ^* \quad [\text{H}, \text{C}] \end{array} $ Exactly 1 hetero atom alpha to C*; 1 electron pair on the phosphorus (or phosphorus connectivity = 3)	Slightly more restrictive in SLICE

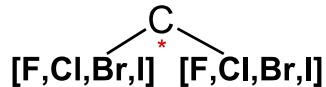
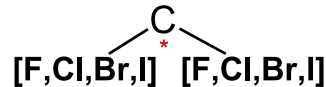
PHOSPHONATE – **rename to o*phosphonate in SLICE or leave it out of SLICE (See the comments)**

CHMTRN/PATRAN	SLICE	Comments
$\begin{array}{c} *C-O \\ \\ P \end{array}$ No H on the oxygen; exactly 1 hetero atom alpha to the oxygen atom	$\begin{array}{c} *C-O \\ \quad // \\ C \quad P=O \\ \quad \\ O-[H,C] \end{array}$ Exactly 1 hetero atom alpha to each of the carbon atoms (when present); carbonyl oxygen connectivity = 1; connectivity for the other oxygen atoms = 2, including H atoms; exactly 1 hetero atom alpha to each oxygen atom	PHOSPHONATE is used in only two Harvard transforms and one SAVI transform (TF 2218). The name is very misleading, since the PATRAN definition covers all sorts of other functional groups as well (e.g. phosphate, phosphinate, groups with sulfur and nitrogen substituents on the phosphorus). I think we should rename it or restrict it to genuine phosphonates and add more functional groups if they are needed (note that there are potentially very many of them if we do that!). I favour restricting it as indicated, because in practice the generic PATRAN definition is not needed. Alternatively we could leave it out of SLICE. Making the change or leaving it out requires only small changes to TF 2218. To be analogous to the naming of other functional groups, such as sulfonates, this one ought to be "O*PHOSPHONATE"

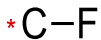
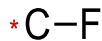
TRIHALIDE

CHMTRN/PATRAN	SLICE	Comments
$\begin{array}{c} [F,Cl,Br,I] \\ \\ *C \\ / \quad \backslash \\ [F,Cl,Br,I] \quad [F,Cl,Br,I] \end{array}$ No hetero atoms alpha to any of the halogen atoms	$\begin{array}{c} [F,Cl,Br,I] \\ \\ *C \\ / \quad \backslash \\ [F,Cl,Br,I] \quad [F,Cl,Br,I] \end{array}$ No hetero atoms alpha to any of the halogen atoms; exactly 3 hetero atoms alpha to C*	More restrictive in SLICE

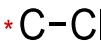
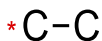
GEM*DIHALIDE

CHMTRN/PATRAN	SLICE	Comments
 No trihalide or acid*halide on C*; no hetero atoms alpha to any of the halogen atoms	 Exactly 2 hetero atoms alpha to C*; C* not multiply bonded; connectivity of each halogen atom = 1	More restrictive in SLICE

FLUORIDE

CHMTRN/PATRAN	SLICE	Comments
 No trihalide, gem*dihalide or acid*halide on C*	 If C* is not aromatic then not multiply bonded and exactly 1 hetero atom alpha to it	More restrictive in SLICE

CHLORIDE

CHMTRN/PATRAN	SLICE	Comments
	 If C* is not aromatic then not multiply bonded and exactly 1 hetero atom alpha to it	More restrictive in SLICE

BROMIDE

#		
CHMTRN/PATRAN	SLICE	Comments
*C-Br	*C-Br If C* is not aromatic then not multiply bonded and exactly 1 hetero atom alpha to it	More restrictive in SLICE

IODIDE

CHMTRN/PATRAN	SLICE	Comments
*C-I	*C-I If C* is not aromatic then not multiply bonded and exactly 1 hetero atom alpha to it	More restrictive in SLICE

HALOAMINE

CHMTRN/PATRAN	SLICE	Comments
$\text{*C-N}\begin{matrix} \diagup \\ \diagdown \end{matrix} \begin{matrix} \text{[F,Cl,Br,I]} \end{matrix}$ Exactly 1 hetero atom alpha to C*; exactly 1 hetero atom alpha to the nitrogen; exactly 1 hetero atom alpha to the halogen	$\text{*C-N}\begin{matrix} \diagup \\ \diagdown \end{matrix} \begin{matrix} \text{[F,Cl,Br,I]} \end{matrix}$ Exactly 1 hetero atom alpha to C*; exactly 1 hetero atom alpha to the nitrogen; exactly 1 hetero atom alpha to the halogen	I.e. no change

CARBONIUM

CHMTRN/PATRN	SLICE	Comments
C^{+*}	C^{+*}	This has never been used in a pattern-based transform in the Harvard database – only in much earlier group-based transforms. It seems unlikely to be needed in SLICE and so it could be dropped.

VIC*DIHALIDE

CHMTRN/PATRN	SLICE	Comments
$ \begin{array}{c} [\text{C},\text{F},\text{Cl},\text{Br},\text{I}] \\ \\ \text{C}^{*}-\text{C} \\ \\ [\text{C},\text{F},\text{Cl},\text{Br},\text{I}] \end{array} $	$ \begin{array}{c} [\text{C},\text{F},\text{Cl},\text{Br},\text{I}] \\ \\ \text{C}^{*}-\text{C} \\ \\ [\text{C},\text{F},\text{Cl},\text{Br},\text{I}] \end{array} $ Exactly 1 hetero atom alpha to each carbon atom Neither carbon atom multiply bonded	Slightly more restrictive in SLICE

AZIDE

CHMTRN/PATRAN	SLICE	Comments
$\begin{array}{c} \text{N}=\text{N}=\text{N}^- \\ \\ \text{*C} \end{array}$ No H alpha to any of the nitrogens; exactly 1 hetero atom alpha to the first and third nitrogen atoms; eExactly 2 hetero atoms alpha to the second nitrogen atom	$\begin{array}{c} \text{N}=\text{N}=\text{N}^- \\ \\ \text{*C} \end{array}$ No H alpha to any of the nitrogens; exactly 1 hetero atom alpha to the first and third nitrogen atoms; exactly 2 hetero atoms alpha to the second nitrogen atom	I.e. no change

DISULFIDE

CHMTRN/PATRAN	SLICE	Comments
$\begin{array}{c} \text{S} \quad \text{S} \\ / \quad \backslash \\ \text{*C} \quad \text{C} \end{array}$ Exactly 1 hetero atom alpha to each carbon; exactly 1 hetero atom alpha to each sulfur atom; 2 electron pairs on each sulfur atom	$\begin{array}{c} \text{S} \quad \text{S} \\ / \quad \backslash \\ \text{*C} \quad \text{C} \end{array}$ Exactly 1 hetero atom alpha to each carbon; 2 electron pairs on each sulfur atom (or connectivity = 2)	I.e. no change

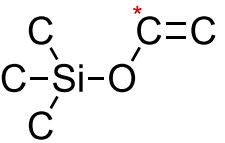
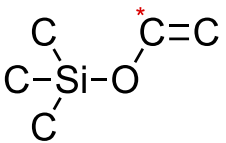
HALOHYDRIN

CHMTRN/PATRAN	SLICE	Comments
Substituents on the oxygen restricted to H, alkyl, acyl.	Exactly 1 hetero atom alpha to C*; exactly 1 hetero atom alpha to C1; C* and C1 not multiply bonded	More restrictive in SLICE The PATRAN allows the oxygen to be OH, OR, or OC(=O)R. This is not consistent with the usual meaning of "halohydrin" which allows only OH. HALOHYDRIN is used in only three, early, group-based transforms (i.e. not pattern based). At least one of them is limited to a "true" halohydrin. So I suggest making that the definition.

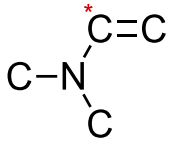
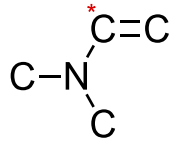
ENOL*ETHER

CHMTRN/PATRAN	SLICE	Comments
Exactly 1 hetero atom alpha to C*; C* and C2 not aromatic; C1 can be ether or ester but not acetal or hemiacetal	Exactly 1 hetero atom alpha to C*; C* and C2 not aromatic; exactly 1 hetero atom alpha to C1	Slightly more restrictive in SLICE

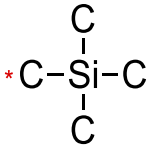
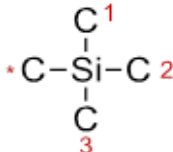
SILYLENOETHER

CHMTRN/PATRAN	SLICE	Comments
 No H on the oxygen atom; doubly bonded carbons not aromatic	 Exactly 1 hetero atom alpha to each of the silicon carbons; no H on the oxygen atom; doubly bonded carbons not aromatic	Slightly more restrictive in SLICE

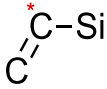
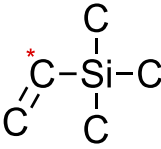
ENAMINE

CHMTRN/PATRAN	SLICE	Comments
 Doubly bonded carbons not aromatic; no H and no hetero atom alpha to the nitrogen atom	 Doubly bonded carbons not aromatic; no H and no hetero atom alpha to the nitrogen atom	I.e. no change

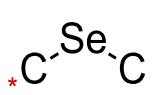
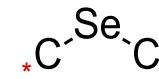
TRIALKYLSILYL

CHMTRN/PATRAN	SLICE	Comments
 <chem>*C[Si](C)C</chem>	 Exactly 1 hetero atom alpha to each of carbon atoms 1, 2, and 3 If C* is not aromatic then exactly one hetero atom alpha to it	Slightly more restrictive in SLICE Small further restriction added

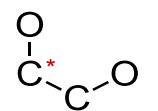
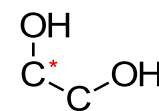
VINYLSILANE

CHMTRN/PATRAN	SLICE	Comments
 Trialkylsilyl group on C*	 Exactly 1 hetero atom alpha to each of the silicon carbons	Slightly more restrictive in SLICE

SELENIDE

CHMTRN/PATRAN	SLICE	Comments
 2 electron pairs on the selenium atom; no hetero atom alpha to the selenium atom	 Exactly 1 hetero atom alpha to each of the carbon atoms; 2 electron pairs on the selenium atom (or connectivity = 2)	More restrictive in SLICE

GLYCOL

CHMTRN/PATRAN	SLICE	Comments
 Groups on the carbon atoms can be alcohol, ether, or ester but not enol ether; oxygen not in rings size < 3	 Exactly 1 hetero atom alpha to each of the carbon atoms; Connectivity = 4 for each carbon atom	More restrictive in SLICE Glycol is not used in any Harvard pattern chemistry transforms

Named generic functional groups

CARBONYL

CHMTRN/PATRAN	SLICE	Comments
ALDEHYDE OR KETONE	Aldehyde or ketone	I.e. no change

CARBOXYL

CHMTRN/PATRAN	SLICE	Comments
ACID, ESTER, AMIDE, ACID*HALIDE, THIOESTER, OR ANHYDRIDE	Acid, ester, amide, acid*halide, thioester, or anhydride	I.e. no change

HALIDE

CHMTRN/PATRAN	SLICE	Comments
FLUORIDE, CHLORIDE, BROMIDE, OR IODIDE	Fluoride, chloride, bromide, or iodide	I.e. no change

AMINE

CHMTRN/PATRAN	SLICE	Comments
AMINE*1, AMINE*2, OR AMINE*3	Amine*1, amine*2, or amine*3	I.e. no change

AMIDE

CHMTRN/PATRAN	SLICE	Comments
AMIDE*1, AMIDE*2, OR AMIDE*3	Amide*1, amide*2, or amide*3	I.e. no change

WITHDRAWING

CHMTRN/PATRAN	SLICE	Comments
KETONE, ALDEHYDE, ESTER, AMIDE, CARBONIUM, ACID*HALIDE, THIOESTER, OXIME, IMINE, NITRILE, NITRO, SULFOXIDE, SULFONE, C*SULPHONATE, ANHYDRIDE, TRIHALIDE, OR VINYLW	Ketone, aldehyde, carboxylic acid, ester, amide, carbonium, acid*halide, thioester, oxime, imine, nitrile, nitro, sulfoxide, sulfone, c*sulphonate, anhydride, trihalide, or vinylw(?)	Carboxylic acid added to the list – its absence in CHMTRN appears to have been a mistake because it was appended, probably later, to the list for, e.g., VINYLW. Inclusion of VINYLW in CHMTRN is (deliberately) potentially recursive in practice (see definition of VINYLW below). Can we support that easily? It would be good if we could but, if not, leave it out or redefine vinylw.

EXPANDABLE*WITHDRAWING

CHMTRN/PATRAN	SLICE	Comments
KETONE, ALDEHYDE, ESTER, AMIDE, CARBONIUM, ACID*HALIDE, THIOESTER, OXIME, IMINE, NITRILE, ANHYDRIDE, OR TRIHALIDE	Ketone, aldehyde, ester, amide, carbonium, acid*halide, thioester, oxime, imine, nitrile, anhydride, or trihalide	I.e. no change

NONEXPANDABLE*WITDRAWING

CHMTRN/PATRAN	SLICE	Comments
NITRO, SULFOXIDE, SULFONE, OR C*SULPHONATE	Nitro, sulfoxide, sulfone, or c*sulphonate	I.e. no change

DONATING

CHMTRN/PATRAN	SLICE	Comments
AMINE, HALOAMINE, HYDROXYLAMINE, THIOL, SULFIDE, PHOSPHINE, ETHER, ALCOHOL, HEMIACETAL, ACETAL, AMIDZ, ESTERX, SELENIDE, TRIALKYLSILOXY, N*UREA, N*CARBAMATE, O*CARBAMATE, O*CARBONATE, OR VINYL D	Amine, haloamine, hydroxylamine, thiol, sulfide, phosphine, ether, alcohol, hemiacetal, acetal, amidz, esterx, selenide, trialkylsiloxo, n*urea, n*carbamate, o*carbamate, o*carbonate, or vinyl d(?)	Inclusion of VINYL D in CHMTRN is (deliberately) potentially recursive (see definition of VINYL D below). Can we support that easily? It would be good if we could but, if not, leave it out or redefine vinyl d.

LEAVING

CHMTRN/PATRAN	SLICE	Comments
AZIRIDINE, THIOCYANATE, THIOL, EPISULFIDE, SULFIDE, SULFONE, C*SULFONATE, O*SULFONATE, PHOSPHONATE, EPOXIDE, ETHER, ALCOHOL, FLUORIDE, CHLORIDE, BROMIDE, IODIDE, GEM*DIHALIDE, TRIHALIDE, ACETAL, ESTERX, AMIDZ, SELENIDE, TRIALKYLSILOXY, N*UREA*H, N*UREA*C, O*CARBAMATE, N*CARBAMATE, O*CARBONATE, DITHIOACETAL, OR DITHIOKETAL	Aziridine, thiocyanate, thiol, episulfide, sulfide, sulfone, c*sulfonate, o*sulfonate, phosphonate, epoxide, ether, alcohol, fluoride, chloride, bromide, iodide, gem*dihalide, trihalide, acetal, esterx, amidz, selenide, trialkylsiloxo, n*urea*h, n*urea*c, o*carbamate, n*carbamate, o*carbonate, dithioacetal, or dithioacetal	I.e. no change

GOOD*LEAVING

CHMTRN/PATRAN	SLICE	Comments
AZIRIDINE, EPISULFIDE, O*SULFONATE, PHOSPHONATE, EPOXIDE, CHLORIDE, BROMIDE, IODIDE, GEM*DIHALIDE, OR TRIHALIDE	Aziridine, episulfide, o*sulfonate, phosphonate, epoxide, chloride, bromide, iodide, gem*dihalide, or trihalide	I.e. no change

VINYLW

CHMTRN/PATRAN	SLICE	Comments
*C=C 1 Withdrawing group on C1; carbon atoms not aromatic	*C=C 1 Withdrawing group on C1; carbon atoms not aromatic	I.e. no change See the comments for WITHDRAWING

VINYLD

CHMTRN/PATRAN	SLICE	Comments
*C=C 1 Donating group on C1; carbon atoms not aromatic	*C=C 1 Donating group on C1; carbon atoms not aromatic	I.e. no change See the comments for DONATING

FUNCTIONAL

CHMTRN/PATRAN	SLICE	Comments
Any functional group	Any functional group	I.e. no change but probably not needed

Substructures that are defined and named but are not functional groups

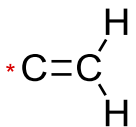
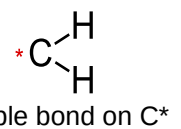
METHYL

CHMTRN/PATRAN	SLICE	Comments
$\begin{array}{c} \text{H} \\ \\ * \text{C} - \text{H} \\ \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} \\ \\ * \text{C} - \text{H} \\ \\ \text{H} \end{array}$	METHYL and METHYL*ATOM (see following item) behave like HYDROGEN and HYDROGEN*ATOM in CHMTRN. This behaviour is not needed in SLICE and the distinction is not necessary to any existing Harvard transforms. So I suggest that the origin of METHYL should be its own carbon, like it is for other functional groups.

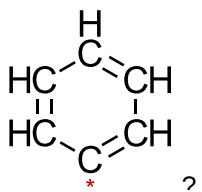
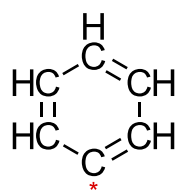
METHYL*ATOM

CHMTRN/PATRAN	SLICE	Comments
$\begin{array}{c} \text{H} \\ \\ * \text{C} - \text{H} \\ \\ \text{H} \end{array}$	Omit	See above

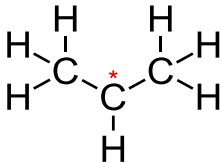
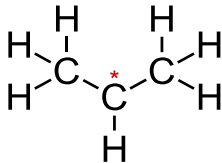
METHYLENE

CHMTRN/PATRAN	SLICE	Comments
	Omit from SLICE or change to:  Double bond on C*	The substructure shown is the one described in the CHMTRN manual but METHYLENE is used only twice in Harvard pattern transforms. In both cases it is used in mechanism statements to attach CH2 to a hetero atom. So either the manual is wrong or the transforms are erroneous. Defining it the way it is described in the manual is confusing and not what one would expect.

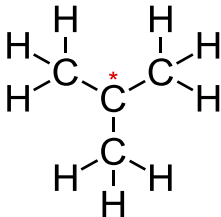
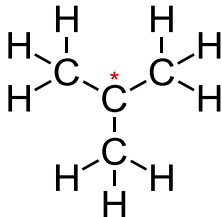
PHENYL

CHMTRN/PATRAN	SLICE	Comments
		I.e. No change except that, because of how it is used in practice in Harvard transforms it is not clear whether the origin is as shown. The ring is aromatic – I gave up trying to find a way to show that explicitly using Chemdoodle! Phenyl is only used in mechanism commands in CHMTRN but I think its use in qualifiers is supported in the current version of SAVI

ISOPROPYL

CHMTRN/PATRAN	SLICE	Comments
		Isopropyl is only used in mechanism commands in CHMTRN but I think its use in qualifiers is supported in the current version of SAVI

T*BUTYL

CHMTRN/PATRAN	SLICE	Comments
		t-Butyl is only used in mechanism commands in CHMTRN but I think its use in qualifiers is supported in the current version of SAVI

ALKYL and ALKYL*ATOM

There is a technical problem with alkyl because, the way chemists use the word, its meaning is context-dependent. In C-R there are no hetero atoms alpha to the first carbon of the alkyl attachment whereas in N-R there is one hetero atom alpha to it. The Harvard CHMTRN interpreter understood this if you asked about, e.g., "alpha to atom*1", where atom 1 is the C or N in C-R or N-R. However, if you asked about atom*2, where that is the first carbon in the R appendage, R would not be recognised as alkyl in the case where it is attached to N. In programming terms this may be logical but it is confusing, especially to someone not familiar with the niceties of program code. A further weakness is that, to a chemist, "alkyl" means "an appendage containing only carbon and hydrogen" (perhaps further limited to there being no multiple bonds), whereas in CHMTRN it means only that there are no heteroatoms attached to the first carbon in the appendage.

I've looked at where ALKYL and ALKYL*ATOM are used in transforms and, given the Harvard definition, they can easily be replaced by asking about hetero atoms attached to the relevant carbon. This both avoids the need for these oddly defined CHMTRN words and simplifies the work of the interpreter program. So I suggest we leave them out of SLICE

Note on use of “PRIMARY”, “SECONDARY”, “TERTIARY”

CHMTRN supports the use of these terms for functional group origins, where relevant.

For example

